

API Testing

Introduction

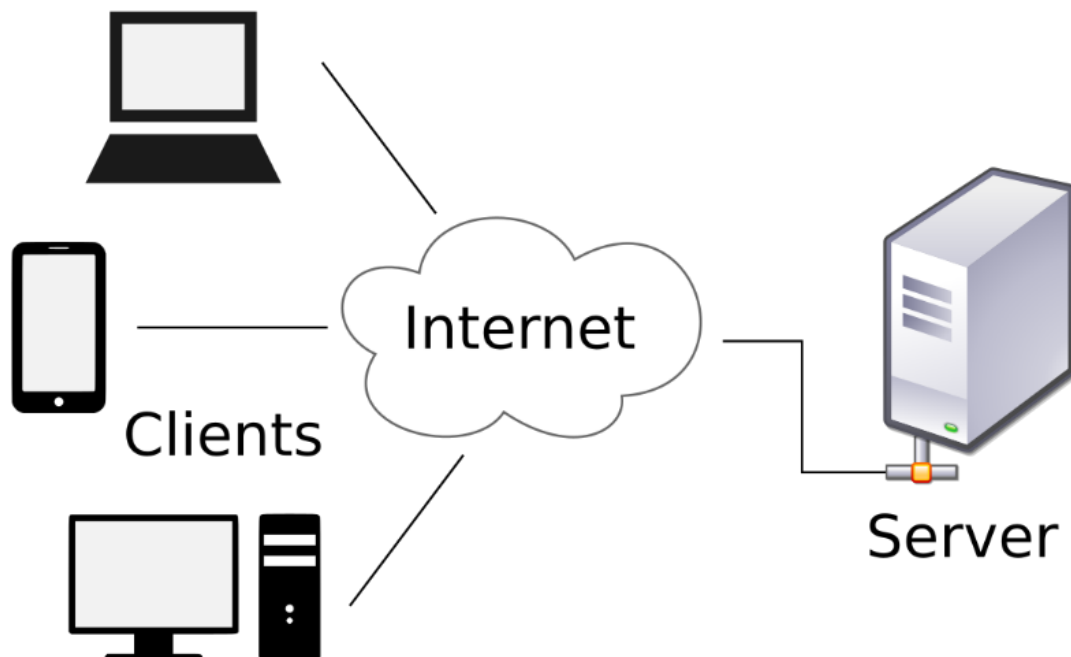
What is Client & Server?

Client is a device or software (like a computer, phone, or browser) that sends requests to get services or data.

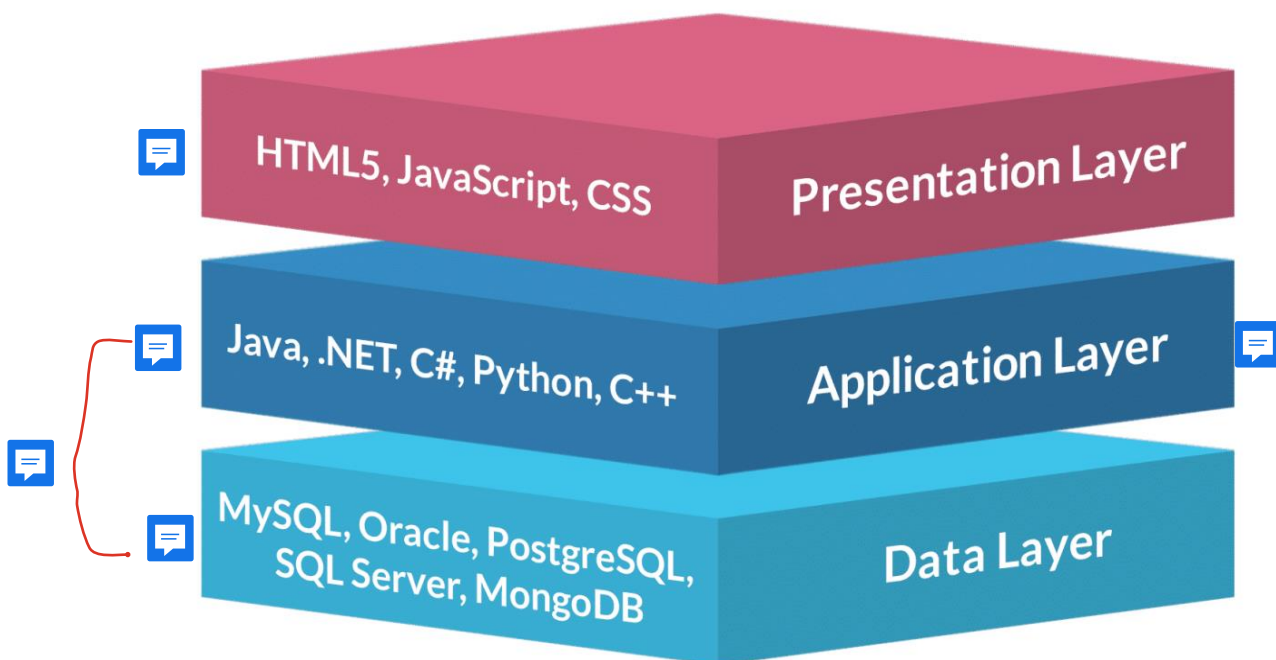
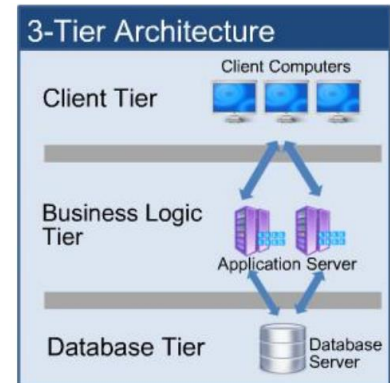
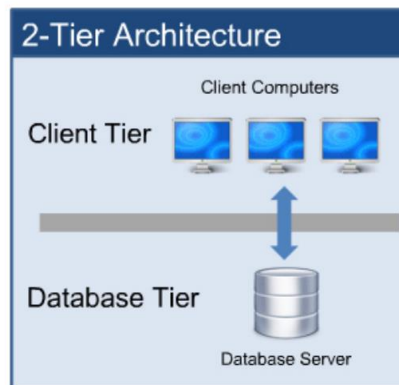
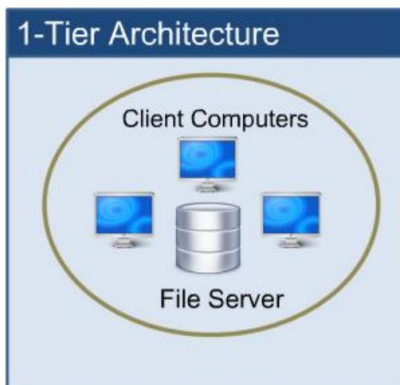
For example, when you use a browser to open a website, your browser acts as the client.

Server is a powerful computer or software that receives those requests, processes them, and sends back the requested data or service.

For instance, when you open a website, the server provides the webpage to your browser.



Client/Server Architecture



What is an API?

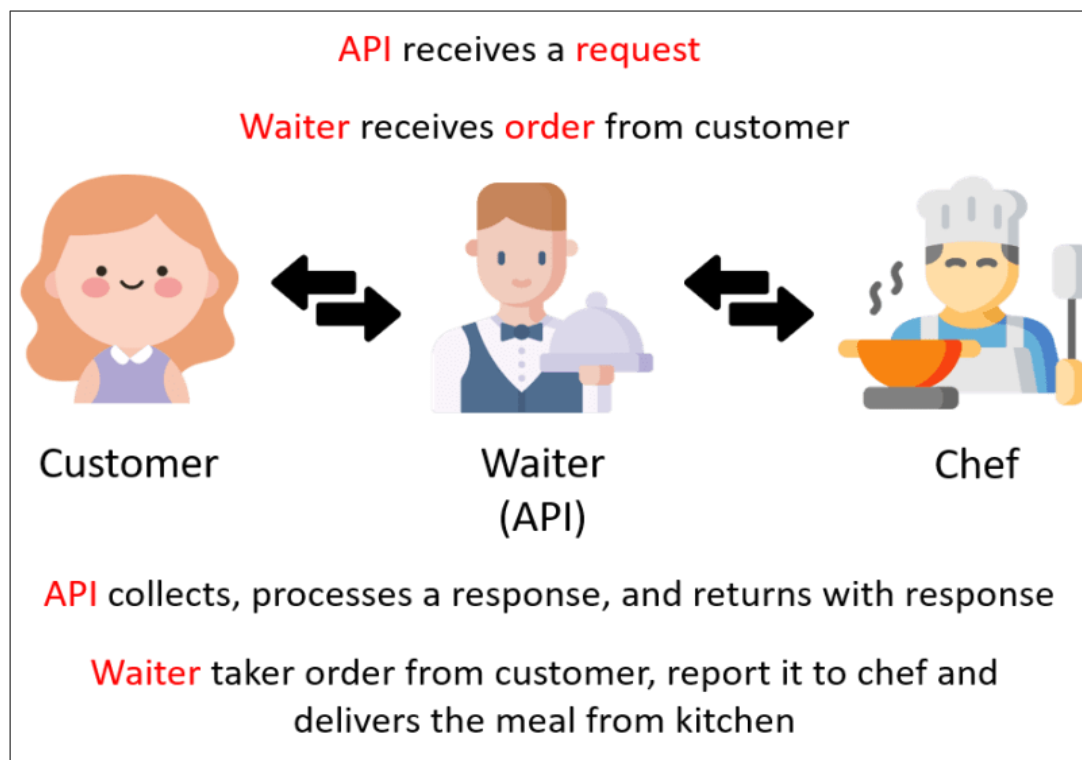
API (Application Programming Interface) is like a messenger that helps two programs or systems talk to each other and share information.

For example:

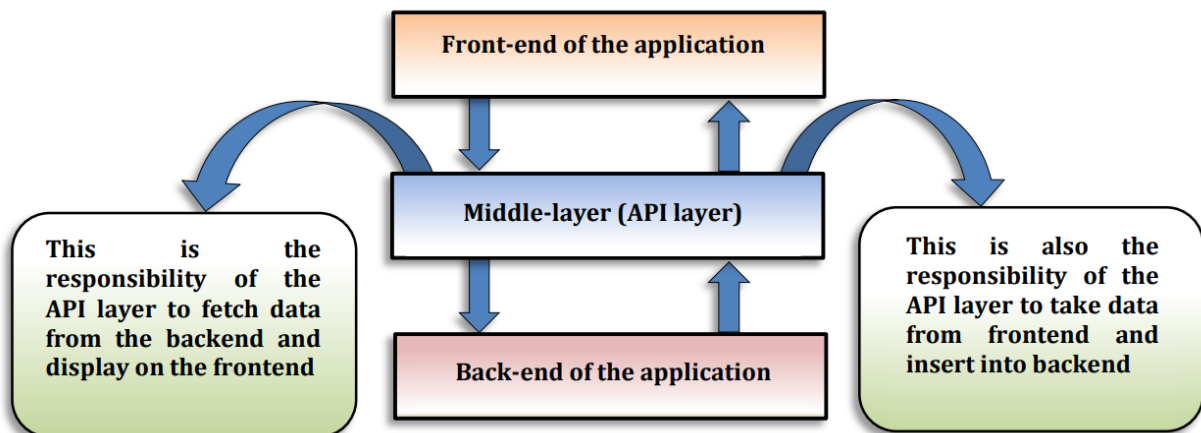
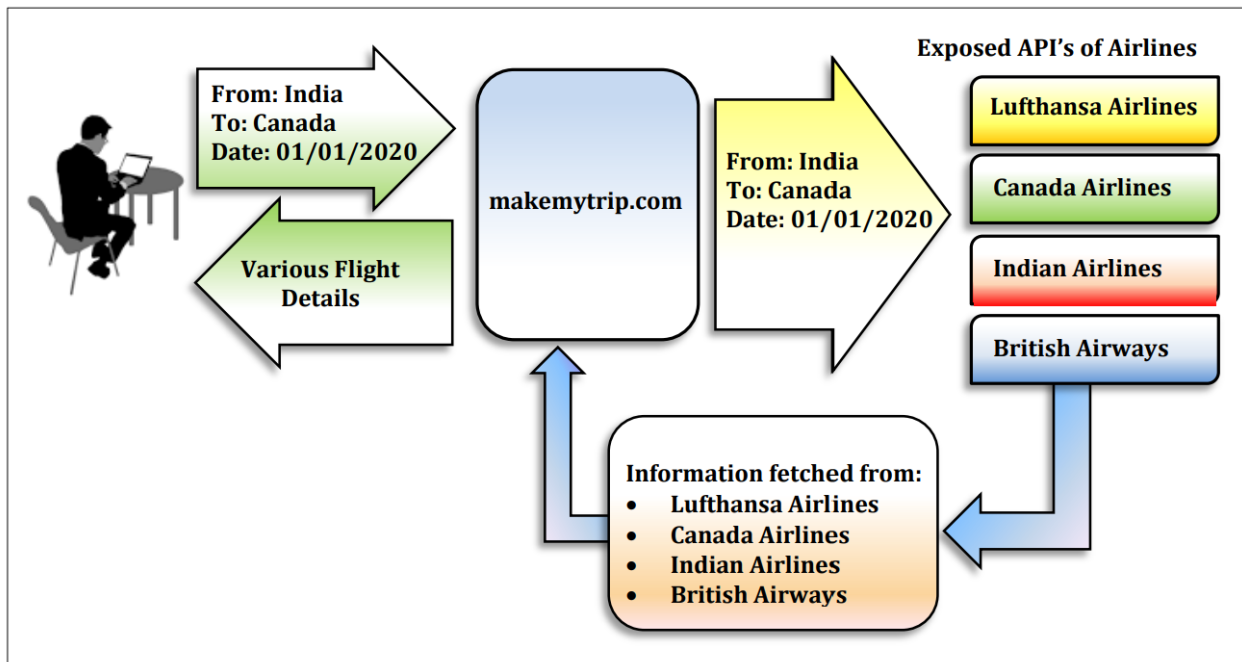
- Imagine you're using a **food delivery app**. When you select a restaurant and place an order, the app uses an API to communicate with the restaurant's system to check the menu, place your order, and confirm it.
- APIs define the rules for how this communication happens, like what data can be sent, how to ask for it, and what the response will look like.

It's **like a waiter in a restaurant**, taking your request to the kitchen and bringing back your food. The API (waiter) ensures both sides (you and the kitchen) understand each other!

API - Restaurant Analogy



API - MakeMyTrip Analogy



**** API is a way of communication between two layers.**

Types Of API

APIs come in different types based on how and where they are used.

Open APIs (Public APIs):

- These are open for everyone to use. Developers can access them without restrictions (or with minimal requirements).
- Example: Google Maps API lets any app show maps and directions.

Internal APIs (Private APIs):

- These are used only within a company or organization. They help internal systems or teams communicate securely.
- Example: A company's HR system API to access employee data for payroll processing.

Partner APIs:

- These are shared with specific partners or businesses. They require special permissions or agreements to use.
- Example: A travel app using an airline's API to show flight details.

Composite APIs:

- These combine multiple APIs into one call, allowing you to get data from different sources in a single request.
- Example: An e-commerce app using a composite API to retrieve product info, pricing, and customer reviews all at once.

API Vs Webservice Vs Microservice

Here's a simple comparison of API, Web Service, and Microservice:

1.API (Application Programming Interface)

- **What it is:** A way for different software systems or applications to communicate with each other.
- **How it works:** It defines rules for how one app can request data or services from another. APIs don't have to use the internet—they can work locally, too.

- **Example:** Your phone's weather app uses an API to get weather data from a weather server.

2. Web Service

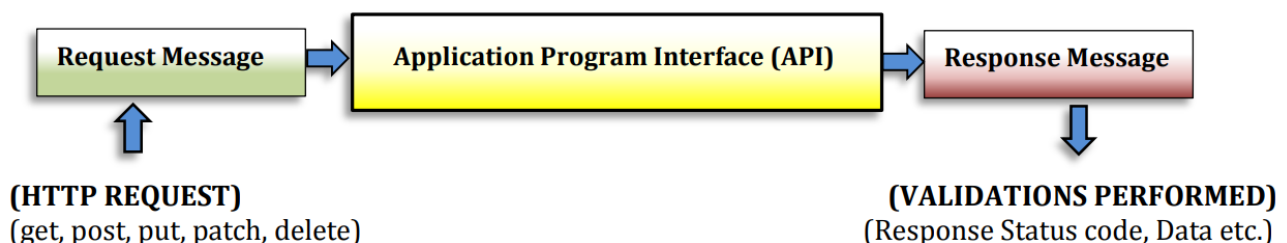
- **What it is:** A type of API that works specifically over the internet (using HTTP).
- **How it works:** It enables two systems to exchange data (often using XML or JSON).
- **Example:** A payment gateway like PayPal's web service allows e-commerce websites to process payments online.

Key point: All web services are APIs, but not all APIs are web services.

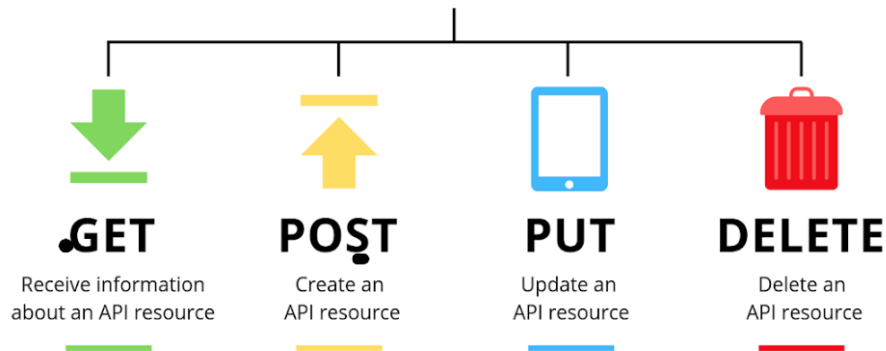
3. Microservice

- **What it is:** A small, independent part of an application that does one thing well. A large app is built by combining many microservices.
- **How it works:** Each microservice communicates with other microservices via APIs. They are self-contained and can run independently.
- **Example:** In an online shopping app, separate microservices might handle user login, product catalogue, payments, and order tracking.

Rest API HTTP Methods



REST API Methods



http Vs https



Helen

HTTP

http://www.example.com

password: abc123



Without password encryption
Hacker see "abc123"



Carol

HTTPS

https://www.example.com

password: abc123



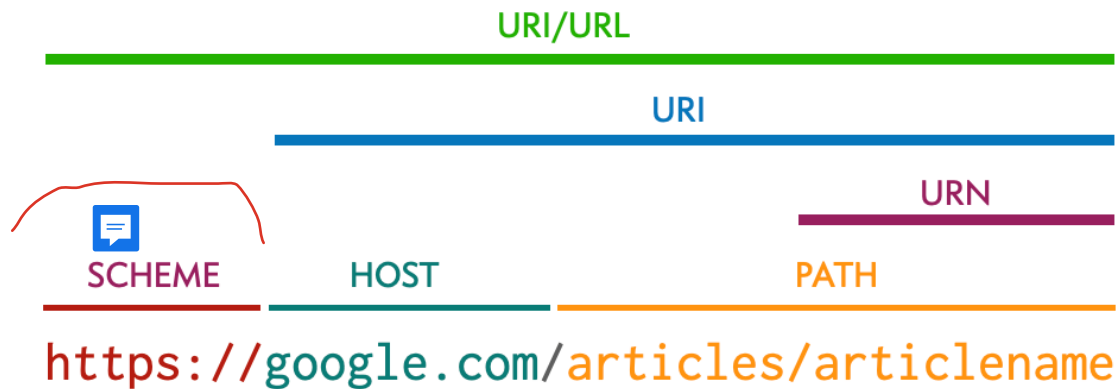
With password encryption
Hacker see "xyaerXzabc"

Terminologies

URI – Uniform Resource Identifier

URL – Uniform Resource Locator

URN – Uniform Resource Name



Feature & Resource

‘**Feature**’ is the term used in manual testing to test some functionality and similarly ‘**Resource**’ is the term used in API Automation testing referring some functionality.

Payload

payload means **body in the HTTP request and response message**.

- Request Payload
- Response Payload

HTTP Status Codes

1XX Informational codes	The server has received the request and is currently processing it
2XX Success codes	The server processed the request after successfully receiving and understanding it
3XX Redirection codes	The server has received the request, but it is being redirected to another location. In some rare instances, additional actions beyond a redirect may be required
4XX Client error codes	The server was unable to locate or access the page or website. This issue originates from the site's end
5XX Server error codes	The client submitted a valid request, but the server was unable to fulfill it

1XX Information		4XX Client (Continue)	
100	Continue	407	Proxy Authentication Required
101	Switching Protocols	408	Request Timeout
102	Processing	409	Conflict
103	Early Hints	410	Gone
2XX Success		411	Length Required
		412	Precondition Failed
200	OK	413	Payload Too Large
201	Created	414	URI Too Large
202	Accepted	415	Unsupported Media Type
203	Non-Authoritative Information	416	Range Not Satisfiable
205	Reset Content	417	Exception Failed
206	Partial Content	418	I'm a teapot
207	Multi-Status (WebDAV)	421	Misdirected Request
208	Already Reported (WebDAV)	422	Unprocessable Entity (WebDAV)
226	IM Used (HTTP Delta Encoding)	423	Locked (WebDAV)
3XX Redirection		424	Failed Dependency (WebDAV)
		425	Too Early
300	Multiple Choices	426	Upgrade Required
301	Moved Permanently	428	Precondition Required
302	Found	429	Too Many Requests
303	See Other	431	Request Header Fields Too Large
304	Not Modified	451	Unavailable for Legal Reasons
305	Use Proxy	499	Client Closed Request
306	Unused	5XX Server Error Responses	
307	Temporary Redirect		
308	Permanent Redirect	500	Internal Server Error
4XX Client Error		501	Not Implemented
		502	Bad Gateway
400	Bad Request	503	Service Unavailable
401	Unauthorized	504	Gateway Timeout
402	Payment Required	505	HTTP Version Not Supported
403	Forbidden	507	Insufficient Storage (WebDAV)
404	Not Found	508	Loop Detected (WebDAV)
405	Method Not Allowed	510	Not Extended
406	Not Acceptable	511	Network Authentication Required
		599	Network Connect Timeout Error